

Problem Statement

Nowadays, public places such as airports, railway stations, shopping malls, schools, and smart cities use CCTV cameras for security and surveillance. However, continuously monitoring multiple camera feeds manually is difficult, time-consuming, and prone to human error. As a result, suspicious activities such as violence, unauthorized access, abandoned objects, crowd panic, or dangerous behavior may not be detected immediately.

This project aims to develop an AI-Based Human Behavior Surveillance and Threat Detection System that can automatically analyze video, image, audio, and text data in real time to identify suspicious human activities and potential threats. The system uses Artificial Intelligence, Computer Vision, and Machine Learning techniques to detect abnormal behavior, aggressive movements, panic situations, and security risks, while generating instant alerts for authorities.

The main objective of this project is to improve public safety, reduce manual monitoring workload, and provide faster emergency response through intelligent automated surveillance.

Proposed Solution

The proposed system uses a multi-modal AI approach to monitor and analyze different types of data such as CCTV video streams, images, emergency audio signals, and security-related text information.

The system first captures real-time surveillance video and processes it using Computer Vision algorithms and object detection models such as YOLO to identify people, suspicious objects, and unusual activities. Human behavior analysis techniques are then applied to recognize activities such as fighting, crowd panic, or unauthorized movement.

Audio analysis models are used to detect emergency sounds such as shouting, screaming, or distress signals. In addition, text-based security alerts or emergency messages can also be analyzed using Natural Language Processing techniques.

All the collected information is combined using multi-modal AI analysis to determine the threat level and generate instant notifications or alerts for security personnel. The system helps improve surveillance efficiency, reduces dependency on manual monitoring, and enables faster response during dangerous situations.